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SAMPLE QUOTATION · EV CHARGER 2ND CIRCUIT · OPTION COMPARISON

Option A vs Option B – EV charger 2nd circuit

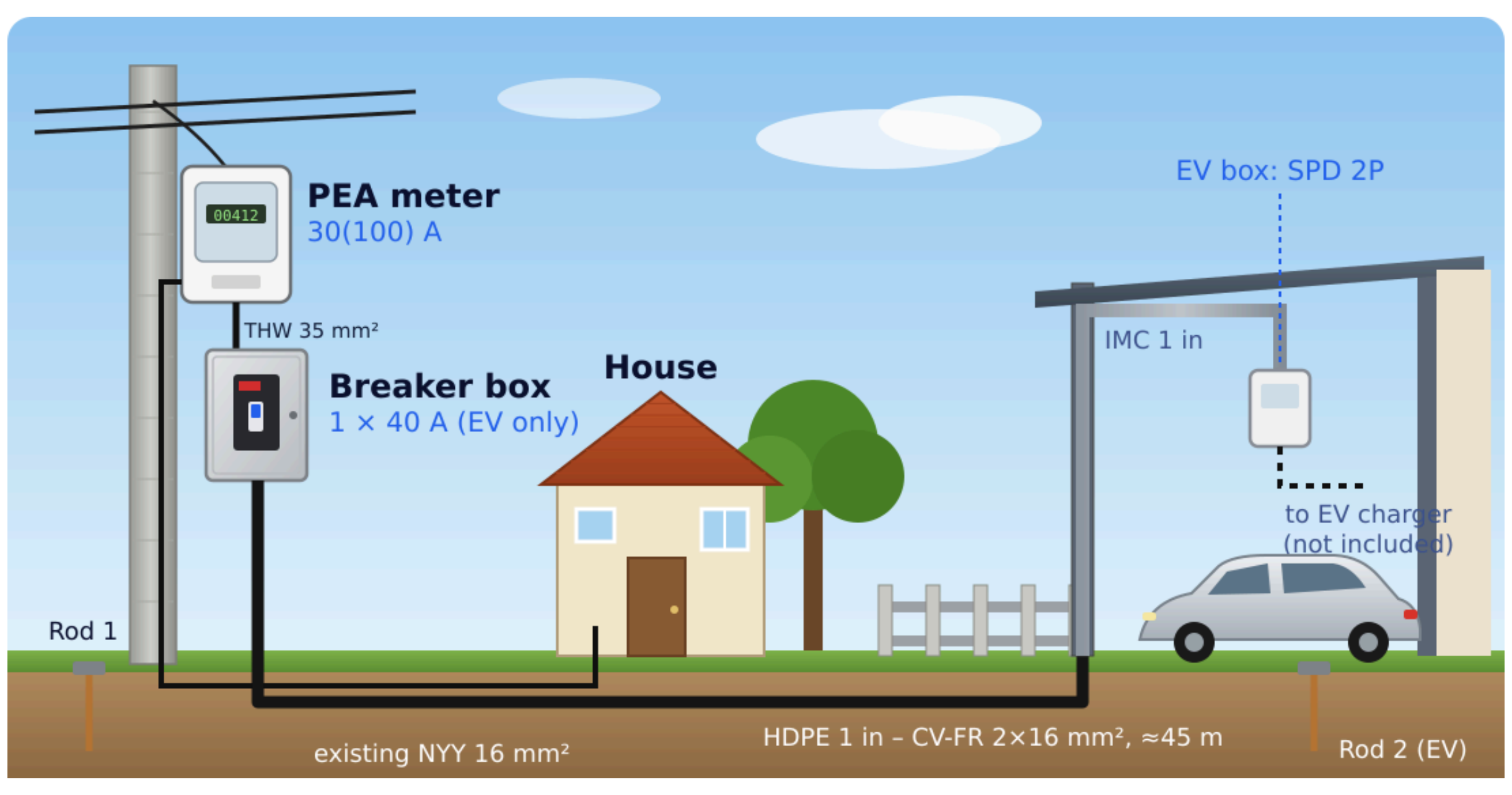
PEA meter 30(100) A · 35 mm² from the meter to the pole box · CV-FR 2 × 16 mm² (≈45 m) in HDPE and IMC 1 in to the EV box · separate EV earth rod (TT) at the EV box · EV charger not included · Nov 2026

Option A

Breaker box at the pole + one 40 A EV breaker

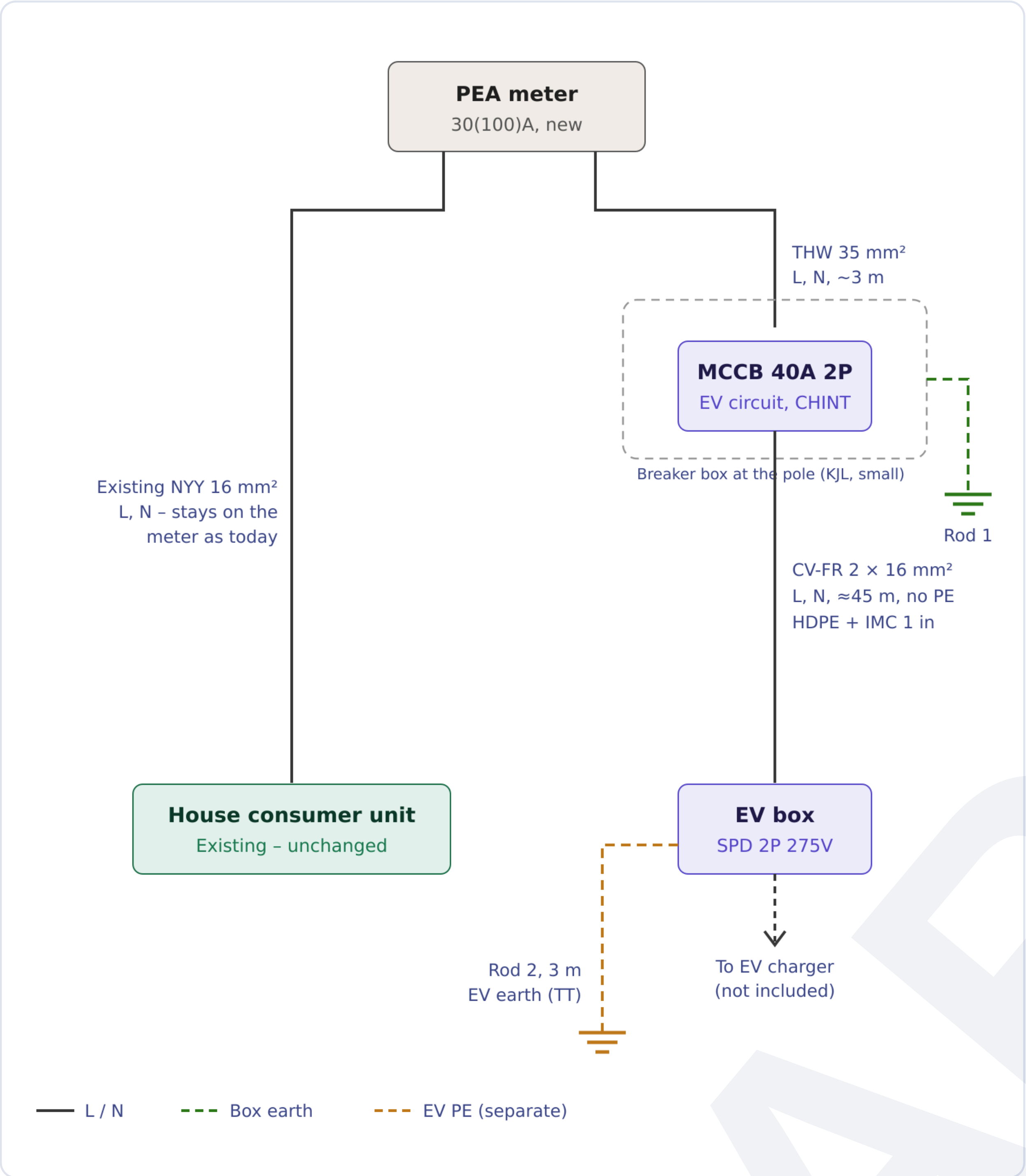
Price on request
ask us for a quotation

1 · Site illustration

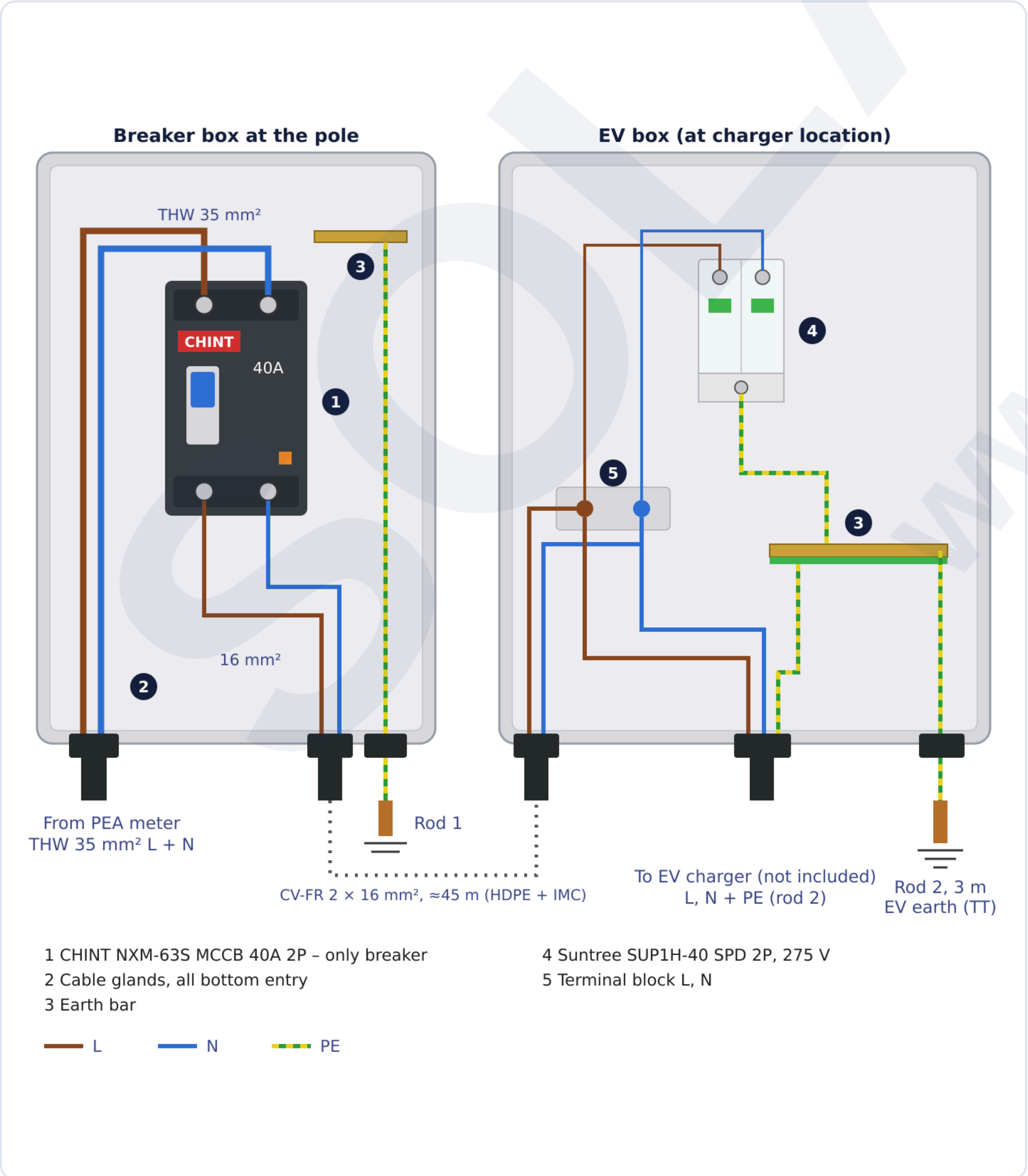

PEA meter
30(100) A
00412
THW 35 mm²
Breaker box
1 × 40 A (EV only)
House
EV box: SPD 2P
IMC 1 in
to EV charger
(not included)
Rod 1
existing NYY 16 mm²
HDPE 1 in – CV-FR 2×16 mm², ≈45 m
Rod 2 (EV)

- Small KJL breaker box at the pole with **one CHINT 40 A 2P breaker** for the EV circuit only
- Fed by **THW 35 mm²** from the PEA meter; EV cable CV-FR 2 × 16 mm² protected from the pole
- EV box with SPD 2P at the charger location; EV earth rod 2 (TT)
- House stays on the meter as today – **no breaker at the pole for the house** and no surge protection for the house

2 · Single-line diagram


PEA meter
30(100)A, new
THW 35 mm²
L, N, ~3 m
MCCB 40A 2P
EV circuit, CHINT
Breaker box at the pole (KJL, small)
Rod 1
Existing NYY 16 mm²
L, N – stays on the
meter as today
CV-FR 2 × 16 mm²
L, N, ≈45 m, no PE
HDPE + IMC 1 in
House consumer unit
Existing – unchanged
EV box
SPD 2P 275V
Rod 2, 3 m
EV earth (TT)
To EV charger
(not included)
L / N
Box earth
EV PE (separate)

3 · Wiring – breaker box and EV box


Breaker box at the pole
EV box (at charger location)
THW 35 mm²
CHINT
40A
16 mm²
1
2
3
4
5
From PEA meter
THW 35 mm² L + N
Rod 1
CV-FR 2 × 16 mm², ≈45 m (HDPE + IMC)
To EV charger (not included)
L, N + PE (rod 2)
Rod 2, 3 m
EV earth (TT)
L
N
PE

1 CHINT NXM-63S MCCB 40A 2P – only breaker
2 Cable glands, all bottom entry
3 Earth bar
4 Suntree SUP1H-40 SPD 2P, 275 V
5 Terminal block L, N

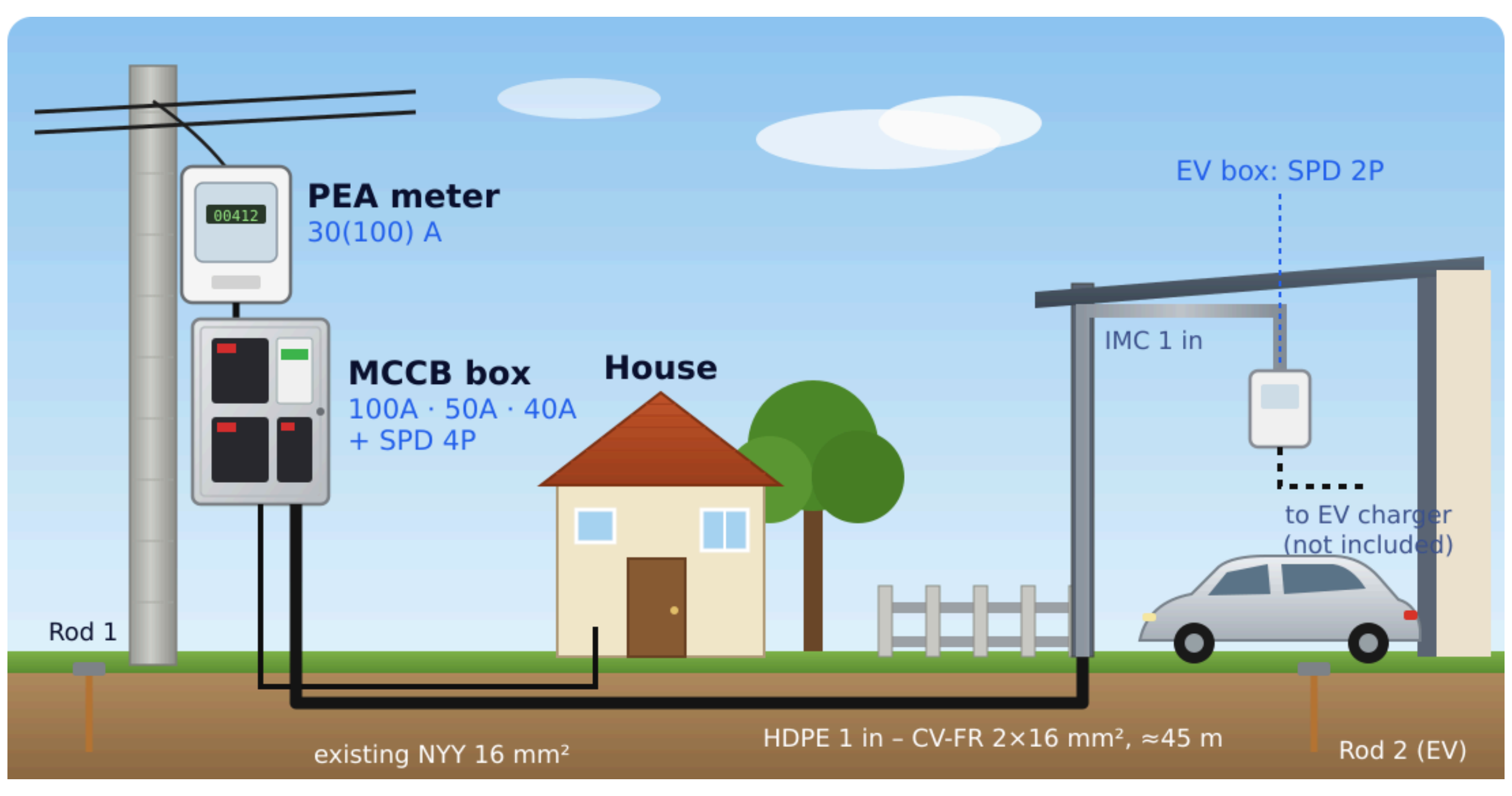
Option B

★ RECOMMENDED

With distribution box – KJL MCCB box at the pole

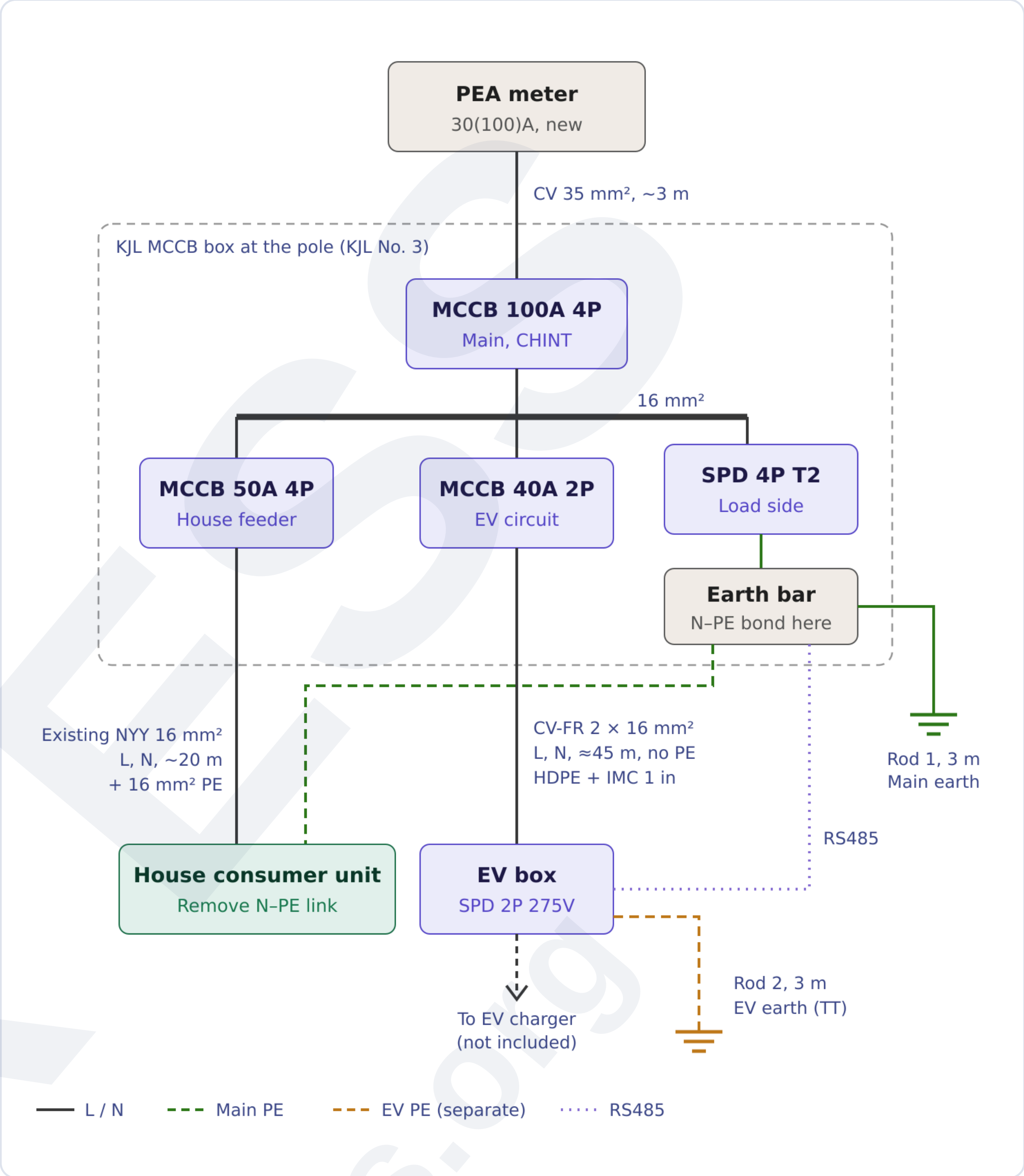
Price on request
ask us for a quotation

1 · Site illustration

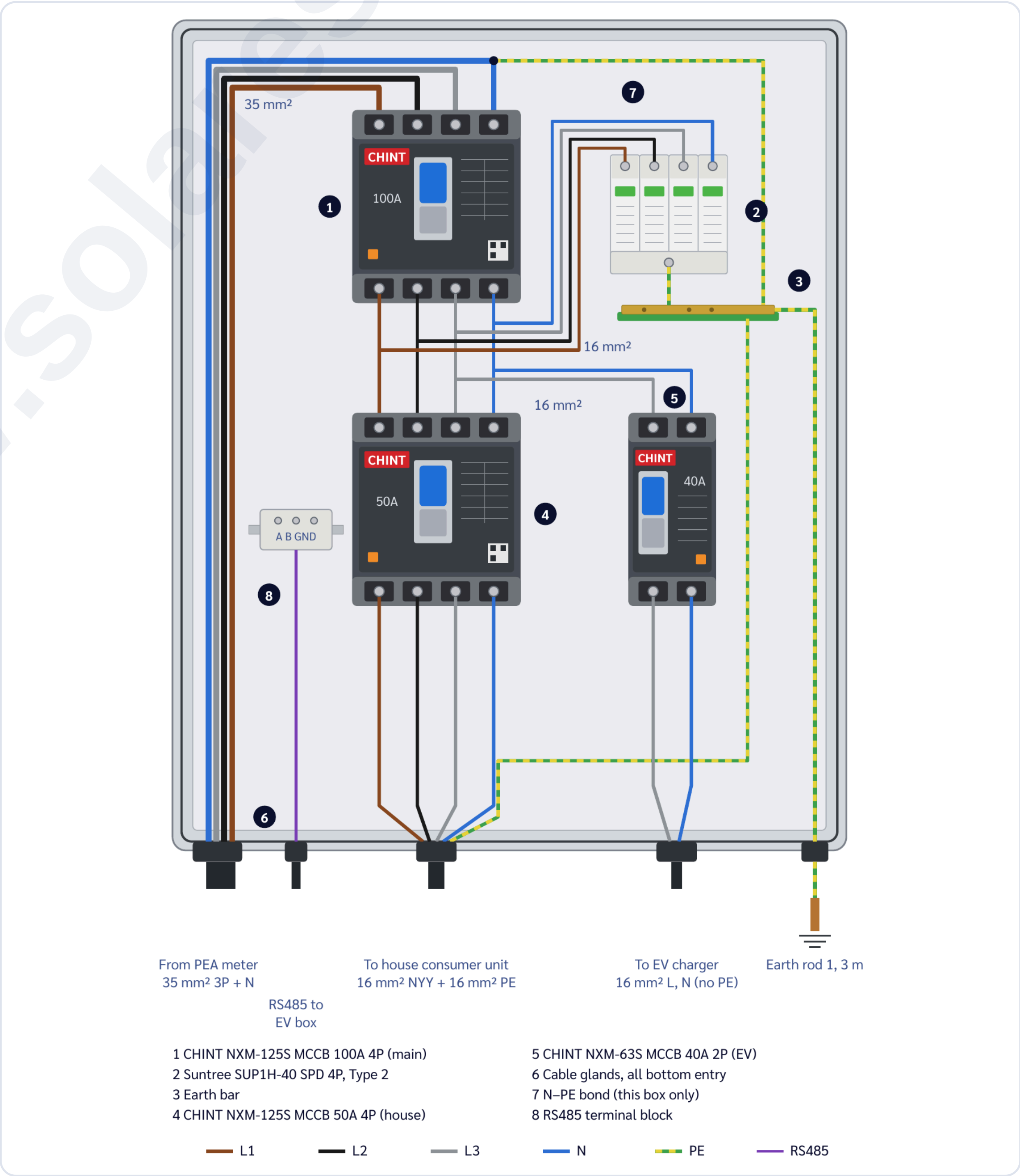

PEA meter
30(100) A
00412
MCCB box
100A · 50A · 40A
+ SPD 4P
House
EV box: SPD 2P
IMC 1 in
to EV charger
(not included)
Rod 1
existing NYY 16 mm²
HDPE 1 in – CV-FR 2×16 mm², ≈45 m
Rod 2 (EV)

- KJL No. 3 box: CHINT **100 A 4P main**, **50 A house feeder**, **40 A EV breaker**
- Suntree **SPD 4P Type 2** for the whole house + SPD 2P in the EV box
- Earth bar with N–PE bond at the box (TN-C-S); RS485 link to the EV box
- **Whole EV cable and house feeder protected** from the pole

2 · Single-line diagram (based on design Rev17)


PEA meter
30(100)A, new
CV 35 mm², ~3 m
KJL MCCB box at the pole (KJL No. 3)
MCCB 100A 4P
Main, CHINT
16 mm²
MCCB 50A 4P
House feeder
MCCB 40A 2P
EV circuit
SPD 4P T2
Load side
Earth bar
N–PE bond here
Existing NYY 16 mm²
L, N, ~20 m
+ 16 mm² PE
CV-FR 2 × 16 mm²
L, N, ≈45 m, no PE
HDPE + IMC 1 in
Rod 1, 3 m
Main earth
RS485
House consumer unit
Remove N–PE link
EV box
SPD 2P 275V
To EV charger
(not included)
Rod 2, 3 m
EV earth (TT)
L / N
Main PE
EV PE (separate)
RS485

3 · Wiring – KJL MCCB distribution box (design Rev17)


35 mm²
CHINT
100A
50A
40A
16 mm²
16 mm²
A B GND
1
2
3
4
5
6
7
8
From PEA meter
35 mm² 3P + N
RS485 to
EV box
To house consumer unit
16 mm² NYY + 16 mm² PE
To EV charger
16 mm² L, N (no PE)
Earth rod 1, 3 m
L1
L2
L3
N
PE
RS485

1 CHINT NXM-125S MCCB 100A 4P (main)
2 Suntree SUP1H-40 SPD 4P, Type 2
3 Earth bar
4 CHINT NXM-125S MCCB 50A 4P (house)
5 CHINT NXM-63S MCCB 40A 2P (EV)
6 Cable glands, all bottom entry
7 N–PE bond (this box only)
8 RS485 terminal block

4 · What each option includes

Item	Option A – 40 A breaker box	Option B – MCCB box
EV package: CV-FR 15 m, HDPE, IMC 1 in, EV box + SPD 2P, 2 earth rods, lugs	✓	✓
PEA meter upgrade to 30(100) A + PEA application	✓	✓
35 mm² cable from the PEA meter to the pole box	THW, L + N	CV 4-core
EV cable CV-FR 2 × 16 mm², ≈45 m in HDPE + IMC	✓	✓
Box at the pole	Small KJL box	KJL No. 3 box
Main breaker MCCB 100 A 4P	–	✓
House feeder breaker MCCB 50 A 4P	–	✓
EV breaker MCCB 40 A 2P	✓	✓
Surge protection SPD 4P Type 2 (whole house)	–	✓
Earth bar with N–PE bond (TN-C-S), RS485 link	–	✓
Installation, testing and handover	✓	✓
EV charger	Not included	Not included

Option B adds
100 A main · 50 A house · SPD 4P
at the pole

Solar ESS recommends Option B.
The house feeder is also switched and protected at the pole, not only the EV circuit, and the whole house gets Type 2 surge protection. Ask us for a quotation for your own site.

The EV charger is not included in either option (owner to supply; it must have built-in RCD Type B, or Type A/F with 6 mA DC detection, as required by PEA/MEA MPESTD-001:2563). Customer details, location and prices have been removed from this sample.

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